

SHS HEAVY SAFE HELICOPTER

SHS-00 | REVISION 19

MASTER ENGINEERING DOCUMENT TREE

Documents 001 – 500 | Sequential Numbering | No Gaps

Baseline: SHS-00 Rev19 | 26,000 kg | Main Rotor 26.0 m | Tail Rotor 4.0 m | Capsule 7,000 kg | 8 persons max | 16 airbags (2 per occupant) | ≥50 cm ergonomic foam seats | 3 primary + 2 reserve canopies | 8 fixed energy-absorbing landing supports

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This Master Document Tree defines the controlled list of engineering documents required to mature SHS Generation 1 from the SHS-00 Revision 19 conceptual baseline toward analysis, design, verification and licensing readiness. Each entry has a sequential number (1–500), document name, and document type. Numbers are continuous with no gaps. Document 001 is SHS-00 Rev19. Roll-cage, sixteen-airbag, ≥50 cm foam seat, staged S0–S5 separation, and 3+2 canopy recovery documents are included and aligned to the Rev19 baseline. Main-rotor separation remains outside the core document scope except as deferred research where explicitly labeled.

Document types include: Master Specification, System Requirements, Architecture Description, Design Description, Interface Control, Analysis Report, Safety Analysis, Test Plan, Test Spec, Mass Budget, Trade Study, IP Description, Licensing Package, Procedure, Checklist, and related controlled types.

Documents 001 – 022 — Program Control, Safety Plans, Mass Top-Level

No.	Document Name	Document Type
1	SHS-00 Comprehensive Engineering Baseline Rev19	Master Specification
2	SHS System Requirements Specification (SyRS)	System Requirements
3	SHS System Architecture Description	Architecture Description
4	SHS Configuration Management Plan	Management Plan
5	SHS Master Document List Control Procedure	Procedure
6	SHS Interface Control Master Index	Interface Control
7	SHS Risk Management Plan	Management Plan
8	SHS Safety Management Plan	Safety Plan
9	SHS Verification and Validation Master Plan	V&V; Plan
10	SHS Certification Basis Definition (CS-29 Oriented)	Certification Basis
11	SHS Concept of Operations (CONOPS)	CONOPS
12	SHS Mission Profile Definition	Mission Definition
13	SHS Design Assurance Level Allocation	Safety Allocation
14	SHS System Safety Assessment Plan	Safety Plan
15	SHS Functional Hazard Assessment (FHA)	Safety Analysis
16	SHS Preliminary System Safety Assessment (PSSA)	Safety Analysis
17	SHS System Safety Assessment (SSA)	Safety Analysis
18	SHS Common Cause Analysis Plan	Safety Analysis
19	SHS Particular Risk Analysis Plan	Safety Analysis
20	SHS Zonal Safety Analysis Plan	Safety Analysis
21	SHS Mass Properties Control Plan	Mass Properties
22	SHS Top-Level Mass Budget 26 000 kg	Mass Budget

Documents 023 – 044 — Structures and Airframe

No.	Document Name	Document Type
23	SHS Component Mass Breakdown Report	Mass Report
24	SHS Center of Gravity Envelope Definition	Mass Properties
25	SHS Inertia Tensor Estimation Report	Mass Properties
26	SHS Structural Design Criteria	Design Criteria
27	SHS Airframe Primary Structure Design Description	Design Description
28	SHS Fuselage Load Path Analysis	Structural Analysis
29	SHS Tail Boom Structural Analysis	Structural Analysis
30	SHS Engine Mount Structure Design	Design Description
31	SHS Transmission Support Structure Design	Design Description
32	SHS Rotor Mast Support Integration Structure	Design Description
33	SHS External Skins and Fairings Design	Design Description
34	SHS Doors Access Panels and Hinges Design	Design Description
35	SHS Firewalls and Structural Fire Barriers	Design Description
36	SHS Structural Fittings Lugs and Fasteners Spec	Specification
37	SHS Cabin Floor Outside Capsule Boundary	Design Description
38	SHS Airframe Fatigue and Damage Tolerance Plan	Structural Analysis
39	SHS Airframe Static Strength Analysis	Structural Analysis
40	SHS Airframe Dynamic Loads Analysis	Structural Analysis

No.	Document Name	Document Type
41	HSH Crashworthiness Criteria Airframe	Design Criteria
42	HSH Materials Selection Report — Airframe	Materials Report
43	HSH Composite Structures Design Allowables	Materials Report
44	HSH Metallic Structures Design Allowables	Materials Report

Documents 045 – 066 — Rotor, Drive, Dynamics

No.	Document Name	Document Type
45	HSH Corrosion Protection and Finishing Spec	Specification
46	HSH Manufacturing Tolerances Airframe	Manufacturing Spec
47	HSH Structural Interface Control Document Cap-AF	Interface Control
48	HSH Ground Handling and Tiedown Provisions	Design Description
49	HSH Maintenance Access and Inspection Provisions	Design Description
50	HSH Structural Weight Optimization Trade Study	Trade Study
51	HSH Main Rotor System Requirements	Requirements Spec
52	HSH Main Rotor 26.0 m Design Description	Design Description
53	HSH Main Rotor Blade Aerodynamic Design	Design Description
54	HSH Main Rotor Blade Structural Design	Design Description
55	HSH Main Rotor Hub and Retention Design	Design Description
56	HSH Main Rotor Mast Design	Design Description
57	HSH Main Rotor Swashplate and Pitch Control	Design Description
58	HSH Main Rotor Pitch Links and Control Rods	Design Description
59	HSH Main Rotor Vibration and Dynamic Control	Design Description
60	HSH Main Rotor RPM and Tip Speed Study	Analysis Report
61	HSH Main Rotor Solidity and Twist Optimization	Trade Study
62	HSH Main Rotor Airfoil Family Selection	Trade Study
63	HSH Main Rotor Disk Loading Performance Study	Performance Analysis
64	HSH Main Rotor Hover Power Required Analysis	Performance Analysis
65	HSH Main Rotor Autorotation Performance Analysis	Performance Analysis
66	HSH Main Rotor Loads Analysis	Structural Analysis

Documents 067 – 088 — Propulsion, Fuel, Thermal

No.	Document Name	Document Type
67	HSH Main Rotor Fatigue Life Assessment	Structural Analysis
68	HSH Main Rotor Ice Protection Concept	Design Description
69	HSH Main Rotor Balancing and Tracking Spec	Specification
70	HSH Tail Rotor System Requirements	Requirements Spec
71	HSH Tail Rotor 4.0 m Design Description	Design Description
72	HSH Tail Rotor Blade Design	Design Description
73	HSH Tail Rotor Hub and Retention Design	Design Description
74	HSH Tail Rotor Gearbox Design	Design Description
75	HSH Tail Rotor Drive Shafting Design	Design Description
76	HSH Tail Rotor Controls and Pitch Mechanism	Design Description
77	HSH Tail Rotor Anti-Torque Authority Analysis	Performance Analysis
78	HSH Tail Rotor Loads and Fatigue Analysis	Structural Analysis
79	HSH Rotor System Interface Control Document	Interface Control
80	HSH Rotor Emergency Release Feasibility Study	Trade Study
81	HSH Propulsion System Requirements	Requirements Spec
82	HSH Turboshift Engine Selection Study	Trade Study
83	HSH Engine Installation Design Description	Design Description
84	HSH Engine Mounts and Vibration Isolation	Design Description
85	HSH Engine Control System Interface Spec	Interface Control
86	HSH Engine Exhaust and Hot Section Installation	Design Description
87	HSH One-Engine-Inoperative Performance Analysis	Performance Analysis
88	HSH Hot and High Performance Analysis	Performance Analysis

Documents 089 – 110 — Performance, Weight & Balance, Propulsion Integration

No.	Document Name	Document Type
89	HSH Main Gearbox Combining Transmission Design	Design Description
90	HSH Intermediate and Reduction Gearbox Design	Design Description
91	HSH Drive Shafts and Couplings Design	Design Description
92	HSH Transmission Lubrication and Cooling System	Design Description
93	HSH Transmission Thermal and Torque Limits Analysis	Analysis Report

No.	Document Name	Document Type
94	HSH Fuel System Requirements Specification	Requirements Spec
95	HSH Fuel Tank Design Crash Fire Resistant	Design Description
96	HSH Fuel Quantity and Usable Volume Analysis	Analysis Report
97	HSH Fuel Pumps Boost and Transfer Design	Design Description
98	HSH Fuel Lines Valves and Fittings Spec	Specification
99	HSH Fuel Isolation and Emergency Shutoff Design	Design Description
100	HSH Fuel Venting Surge and Inerting Provisions	Design Description
101	HSH Unusable Fuel Allocation Report	Mass Report
102	HSH 1000 km Mission Fuel Model	Performance Analysis
103	HSH Fuel System Fire Hazard Analysis	Safety Analysis
104	HSH Engine Fuel Isolation Coordination with Separation	Interface Control
105	HSH Propulsion System Interface Control Document	Interface Control
106	HSH Engine Separation Feasibility Study	Trade Study
107	HSH Transmission Separation Feasibility Study	Trade Study
108	HSH Propulsion Health Monitoring Requirements	Requirements Spec
109	HSH Propulsion Maintenance Concept	Maintenance Concept
110	HSH Propulsion Weight and Balance Contribution	Mass Report

Documents 111 – 133 — Survival Capsule, Roll-Cage, Seats, Airbags, Interfaces

No.	Document Name	Document Type
111	HSH Survival Capsule Requirements Specification	Requirements Spec
112	HSH Capsule Geometry and Layout Definition (8 seats 16 airbags)	Design Description
113	HSH Capsule 7000 kg Mass Budget Bottom-Up (incl. Airbags and Foam Seats)	Mass Budget
114	HSH Capsule Primary Crashworthy Cell Design	Design Description
115	HSH Capsule Internal Roll-Cage CFRP Design	Design Description
116	HSH Capsule Roll-Cage Joints Geometry Arm Cross-Section and Titanium Nodes	Design Description
117	HSH Capsule Outer Shell Fairing Thermal Protection	Design Description
118	HSH Capsule Structural Load Path Analysis	Structural Analysis
119	HSH Capsule Crash Load Analysis	Structural Analysis
120	HSH Capsule Fatigue and Damage Tolerance Analysis	Structural Analysis
121	HSH Capsule Occupant Injury Criteria Definition (Seats Foam Airbags)	Design Criteria
122	HSH Capsule Energy-Absorbing Seats with ≥50 cm Ergonomic Foam Design	Design Description
123	HSH Capsule Seat Restraint System Spec for Thick-Foam Seats	Specification
124	HSH Capsule Floor and Secondary Structure Design	Design Description
125	HSH Capsule Emergency Exits Doors Frames Seals	Design Description
126	HSH Capsule Emergency Egress Analysis (Airbag Compatible Paths)	Safety Analysis
127	HSH Capsule Independent Emergency Power Design	Design Description
128	HSH Capsule Emergency Battery System Spec	Specification
129	HSH Capsule Life Support Oxygen Environmental	Design Description
130	HSH Capsule Fire Smoke Detection Suppression (Airbag Compatible)	Design Description
131	HSH Capsule Water Landing Flotation Sealing	Design Description
132	HSH Capsule Thick-Foam Seat Interface Control CAP-SEAT	Interface Control
133	HSH Capsule Sixteen-Airbag Interface Control CAP-AB	Interface Control

Documents 134 – 156 — Capsule Interfaces, Landing Supports, CAP-LG

No.	Document Name	Document Type
134	HSH Capsule Separation Interface Hardware Design	Design Description
135	HSH Capsule Attachment Fittings and Seat/Airbag Load Path Design	Design Description
136	HSH Capsule-to-Airframe Interface Control CAP-AF	Interface Control
137	HSH Capsule Clearance Verification Requirements	Requirements Spec
138	HSH Capsule Crash Test Plan	Test Plan
139	HSH Capsule Sixteen-Airbag Sled and Impact Test Plan	Test Plan
140	HSH Capsule Materials Selection CFRP Metallic and Airbag Compatibility	Materials Report
141	HSH Capsule Sixteen-Airbag Deployment Logic and Safety Kernel Interface	Design Description
142	HSH Capsule Sixteen-Airbag Inflators Sensors and Wiring Design	Design Description
143	HSH Capsule Growth Allowance Management including Airbag Seat Mass	Mass Report
144	HSH Capsule Sixteen-Airbag System Requirements	Requirements Spec
145	HSH Capsule Sixteen-Airbag Placement and Packaging Design	Design Description
146	HSH Capsule Occupant Protection Test Plan (Seats Foam Airbags)	Test Plan
147	HSH Eight Landing Supports Requirements	Requirements Spec
148	HSH Eight Landing Supports Design Description	Design Description
149	HSH Landing Support Energy Absorber Design	Design Description

No.	Document Name	Document Type
150	HSH Landing Support Load Path Analysis	Structural Analysis
151	HSH Landing Support Asymmetric Failure Analysis	Structural Analysis
152	HSH Landing Support Terrain Interaction Study	Analysis Report
153	HSH Landing Support Ground Clearance Analysis	Analysis Report
154	HSH Landing Support Rotor Downwash Interaction	Analysis Report
155	HSH Landing Support Replaceable Crush Elements Spec	Specification
156	HSH Landing Support Interface Control CAP-LG	Interface Control

Documents 157 – 178 — Recovery Canopies (3+2), Multi-Mode Recovery, Separation Mechanisms

No.	Document Name	Document Type
157	HSH Capsule Parachute System Requirements	Requirements Spec
158	HSH Capsule Recovery Canopy Architecture 3 Primary + 2 Reserve	Design Description
159	HSH Capsule Parachute Sizing Analysis for 7000 kg Capsule	Analysis Report
160	HSH Capsule Parachute Deployment Sequence Design	Design Description
161	HSH Capsule Parachute Reefing and Riser Design	Design Description
162	HSH Capsule Parachute Failure Tolerance Analysis	Safety Analysis
163	HSH Capsule Parachute Inflation Dynamics Study	Analysis Report
164	HSH Capsule Parachute Interface Control CAP-PAR	Interface Control
165	HSH Recovery Canopy Count Trade Study (3+2 vs 4+2 vs 4+4)	Trade Study
166	HSH Multi-Mode Recovery Architecture Description	Architecture Description
167	HSH Controlled Flight Recovery Mode Definition	Design Description
168	HSH Autorotation Recovery Mode Definition	Design Description
169	HSH Selective Mass Hazard Separation Mode	Design Description
170	HSH Capsule Separation Recovery Mode Definition	Design Description
171	HSH Post-Landing Survival Locator Communications	Design Description
172	HSH Rapid Separation System Requirements	Requirements Spec
173	HSH Selective Separation Philosophy Description	Architecture Description
174	HSH State-Dependent Release Logic Specification	Requirements Spec
175	HSH Candidate Separable Components List	Design Description
176	HSH Primary Capsule Release Mechanisms Design	Design Description
177	HSH Backup Independent Release Mechanisms Design	Design Description
178	HSH Release Mechanism Locks and Inhibits Design	Design Description

Documents 179 – 200 — Separation Sensing, Trajectory, Independence, Simulation

No.	Document Name	Document Type
179	HSH Release State Sensing System Design	Design Description
180	HSH Independent Release Power and Signalling	Design Description
181	HSH Release Arm Fire Sequence Specification	Requirements Spec
182	HSH Fuel Energy Isolation Before Release Spec	Requirements Spec
183	HSH Asymmetric Partial Release Detection Design	Design Description
184	HSH Post-Release Trajectory Monitoring Concept	Design Description
185	HSH Capsule Clear Verification Before Parachute	Requirements Spec
186	HSH Failed Release Rollback Fallback Definition	Requirements Spec
187	HSH Staged Separation Timeline S0–S5 Development Spec	Requirements Spec
188	HSH Separation Kinematics Analysis	Analysis Report
189	HSH Separation Trajectory Clearance Analysis	Analysis Report
190	HSH Released Component Collision Envelope Analysis	Safety Analysis
191	HSH Separation System Mass Allocation 1000 kg	Mass Report
192	HSH Separation System Interface Control JET-CMD	Interface Control
193	HSH Separation System Interface Control JET-PWR	Interface Control
194	HSH Separation System Interface Control JET-SNS	Interface Control
195	HSH Separation Common Cause Independence Analysis	Safety Analysis
196	HSH Unintended Separation Hazard Analysis	Safety Analysis
197	HSH Separation Mechanism Reliability Target Spec	Requirements Spec
198	HSH Separation Ground Test Plan	Test Plan
199	HSH Separation Integrated Demonstration Plan	Test Plan
200	HSH Separation Dynamic Simulation Requirements	Requirements Spec

Documents 201 – 226 — Six-Layer Safety, Safety Kernel, State Machine, AI Boundaries

No.	Document Name	Document Type
201	HSH Six-Layer Survival Architecture Description	Architecture Description
202	HSH Layer L1 Automatic Detection Initiation Spec	Requirements Spec

No.	Document Name	Document Type
203	HSH Layer L2 Independent Backup Initiation Spec	Requirements Spec
204	HSH Layer L3 Manual Crew Authority Spec	Requirements Spec
205	HSH Layer L4 Selective Physical Separation Spec	Requirements Spec
206	HSH Layer L5 Direct Impact Survival Spec	Requirements Spec
207	HSH Layer L6 Recovery Post-Event Survival Spec	Requirements Spec
208	HSH Safety Layer Independence Assessment Plan	Safety Analysis
209	HSH Deterministic Safety Kernel Requirements	Requirements Spec
210	HSH Safety Kernel Authorization Preconditions Spec	Requirements Spec
211	HSH Safety Kernel Altitude Airspeed Envelope Logic	Design Description
212	HSH Safety Kernel Attitude Rate Limits Logic	Design Description
213	HSH Safety Kernel Rotor Engine Fire State Logic	Design Description
214	HSH Safety Kernel Structural Health Logic	Design Description
215	HSH Safety Kernel Sensor Validity Disagreement Logic	Design Description
216	HSH Safety Kernel Human Authority Override Spec	Requirements Spec
217	HSH Safety Kernel Software Design Assurance	Software Spec
218	HSH Emergency State Machine Specification	Requirements Spec
219	HSH Emergency State Transition Matrix	Design Description
220	HSH State-Dependent Separation Set Definition	Design Description
221	HSH Intelligent Monitoring Prediction Boundary Spec	Requirements Spec
222	HSH AI Decision Support Constraints Specification	Requirements Spec
223	HSH Health Monitoring System Requirements	Requirements Spec
224	HSH Digital Twin Estimation Concept	Design Description
225	HSH Predictive Fault Detection Concept	Design Description
226	HSH Immutable Emergency Event Record Spec	Requirements Spec

Documents 227 – 250 — Avionics and Flight Control

No.	Document Name	Document Type
227	HSH Avionics System Requirements Specification	Requirements Spec
228	HSH Flight Control Computers Design Description	Design Description
229	HSH Flight Control Actuators Servo Spec	Specification
230	HSH Navigation Air Data Flight Sensors Spec	Specification
231	HSH Communications Mission Avionics Spec	Specification
232	HSH Electrical Generation Distribution Design	Design Description
233	HSH Electrical Wiring Harness Connectors Spec	Specification
234	HSH Battery Emergency Electrical Interfaces	Design Description
235	HSH Hydraulic System Design Description	Design Description
236	HSH Thermal Management Cooling Design	Design Description
237	HSH Environmental Control Ventilation Design	Design Description
238	HSH Fire Detection Warning Suppression Interfaces	Design Description
239	HSH Health Monitoring Data Recording Design	Design Description
240	HSH Lighting Service Systems Design	Design Description
241	HSH Avionics Interface Control AV-CTRL Safety Kernel	Interface Control
242	HSH Electromagnetic Compatibility Plan	EMC Plan
243	HSH Lightning Protection Design	Design Description
244	HSH Software Configuration Integrity Controls	Software Spec
245	HSH Avionics Weight Allocation Report	Mass Report
246	HSH Avionics Failure Modes Effects Analysis	Safety Analysis
247	HSH Avionics Environmental Qualification Plan	Test Plan
248	HSH Human Machine Interface Emergency Controls	Design Description
249	HSH Crew Procedures Emergency Separation Manual	Procedure
250	HSH Training Requirements Emergency Systems	Training Spec

Documents 251 – 275 — Electrical, Hydraulic, Environmental

No.	Document Name	Document Type
251	HSH Performance Requirements Specification	Requirements Spec
252	HSH Power Required Power Available Analysis	Performance Analysis
253	HSH Hover Ceiling Hot High Analysis	Performance Analysis
254	HSH Rate of Climb Cruise Performance Analysis	Performance Analysis
255	HSH Cruise Drag Polar Estimation	Performance Analysis
256	HSH Mission Fuel Flow Analysis 1000 km	Performance Analysis
257	HSH Reserve Policy Definition	Requirements Spec
258	HSH Rotor Vibration Dynamic Stability Analysis	Analysis Report

No.	Document Name	Document Type
259	HSH 6-DOF Emergency Dynamics Model Spec	Requirements Spec
260	HSH 6-DOF Aircraft Pre-Release Dynamics	Analysis Report
261	HSH 6-DOF Capsule Post-Separation Dynamics	Analysis Report
262	HSH 6-DOF Released Component Trajectory Models	Analysis Report
263	HSH 6-DOF Partial Asymmetric Release Cases	Analysis Report
264	HSH 6-DOF Parachute Inflation Transient Loads	Analysis Report
265	HSH 6-DOF Ground Contact Kinematics Slope Terrain	Analysis Report
266	HSH 6-DOF Occupant Acceleration Pulse Screening	Analysis Report
267	HSH 6-DOF Model Uncertainty Sensitivity Report	Analysis Report
268	HSH Flight Envelope Definition Normal Emergency	Requirements Spec
269	HSH Handling Qualities Requirements	Requirements Spec
270	HSH Stability and Control Analysis	Analysis Report
271	HSH Aerodynamic Database Development Plan	Analysis Plan
272	HSH Wind Tunnel Test Plan if Applicable	Test Plan
273	HSH CFD Analysis Plan Rotor Airframe	Analysis Plan
274	HSH Noise Certification Assessment Plan	Certification Plan
275	HSH Performance Flight Test Plan Outline	Test Plan

Documents 276 – 300 — Human Factors, Crew, Cabin Outside Capsule

No.	Document Name	Document Type
276	HSH Operating Fluids Consumables Mass Report	Mass Report
277	HSH Growth Configuration Reserve Management	Mass Report
278	HSH Weight Empty Definition Report	Mass Report
279	HSH Payload Occupancy Accommodation Study	Trade Study
280	HSH Cabin Ergonomics 10-12 Occupants Study	Design Description
281	HSH Geometry Overall Length Width Height Closure	Design Description
282	HSH Ground Clearance Rotor Downwash Study	Analysis Report
283	HSH External Geometry Aerodynamic Fairing Design	Design Description
284	HSH Fixed Landing Support Permanent Drag Impact	Analysis Report
285	HSH Performance Comparison vs Conventional Rotorcraft	Trade Study
286	HSH Hover Endurance Derivation After Power Closure	Performance Analysis
287	HSH OEI Safe Operating Envelope Definition	Requirements Spec
288	HSH Autorotation Entry and Flare Analysis	Performance Analysis
289	HSH Transmission Thermal Limit Performance Impact	Analysis Report
290	HSH Density Altitude Performance Sensitivity	Performance Analysis
291	HSH Flight Manual Performance Section Outline	Documentation Spec
292	HSH Performance Model Validation Plan	V&V; Plan
293	HSH Simulation Facility Requirements 6-DOF	Requirements Spec
294	HSH Pilot-in-the-Loop Simulation Plan Outline	Test Plan
295	HSH Handling Qualities Simulation Matrix	Test Plan
296	HSH Emergency Procedure Simulation Matrix	Test Plan
297	HSH Performance Uncertainty Quantification	Analysis Report
298	HSH Mission Effectiveness Analysis Outline	Analysis Report
299	HSH Operational Suitability Assessment Outline	Analysis Report
300	HSH Performance Closure Gate G1 Checklist	Checklist

Documents 301 – 330 — Simulation, Loads, Dynamics Validation

No.	Document Name	Document Type
301	HSH Verification Matrix Master	V&V; Plan
302	HSH G0 Architecture Freeze Checklist	Checklist
303	HSH G1 Analytical Closure Checklist	Checklist
304	HSH G2 Subsystem Demonstration Plan	Test Plan
305	HSH G3 Integrated Ground Demonstration Plan	Test Plan
306	HSH G4 Dynamic 6-DOF Simulation Plan	Test Plan
307	HSH G5 Structural Crash Testing Plan	Test Plan
308	HSH G6 Recovery System Testing Plan	Test Plan
309	HSH G7 Flight Development Plan Outline	Test Plan
310	HSH G8 Certification Qualification Package Outline	Certification Plan
311	HSH Release Mechanism Subsystem Test Spec	Test Spec
312	HSH Lock Inhibit Subsystem Test Spec	Test Spec
313	HSH Energy Absorber Subsystem Test Spec	Test Spec
314	HSH Parachute Component Test Spec	Test Spec

No.	Document Name	Document Type
315	HSH Sensor Independent Power Test Spec	Test Spec
316	HSH Capsule Structural Static Test Spec	Test Spec
317	HSH Capsule Crash Test Spec	Test Spec
318	HSH Seat Occupant Protection Test Spec	Test Spec
319	HSH Landing Support Drop Test Spec	Test Spec
320	HSH Capsule Attachment Fitting Test Spec	Test Spec
321	HSH Separation Timing Reliability Test Spec	Test Spec
322	HSH Safety Kernel Software Test Spec	Test Spec
323	HSH End-to-End Survival Sequence Ground Test	Test Spec
324	HSH Partial Asymmetric Separation Test Cases	Test Spec
325	HSH Parachute Deployment Inflation Test Spec	Test Spec
326	HSH Canopy Failure Case Test Spec	Test Spec
327	HSH Post-Landing Stability Rescue Access Test	Test Spec
328	HSH Environmental Qualification Test Matrix	Test Plan
329	HSH EMI Lightning Test Plan Outline	Test Plan
330	HSH Fire Fuel Isolation Test Plan Outline	Test Plan

Documents 331 – 360 — Test Plans and Test Specifications

No.	Document Name	Document Type
331	HSH Test Instrumentation Data Acquisition Spec	Specification
332	HSH Test Safety Plan Ground Demonstrations	Safety Plan
333	HSH Flight Test Safety Plan Outline	Safety Plan
334	HSH Evidence Package Structure Definition	Documentation Spec
335	HSH Requirements Traceability Matrix	Traceability Matrix
336	HSH Verification Evidence Register	Register
337	HSH Nonconformance and Deviation Procedure	Procedure
338	HSH Test Report Template Standard	Template
339	HSH Analysis Report Template Standard	Template
340	HSH Design Review Entry Exit Criteria	Procedure
341	HSH Preliminary Design Review Plan	Review Plan
342	HSH Critical Design Review Plan	Review Plan
343	HSH Test Readiness Review Plan	Review Plan
344	HSH Flight Readiness Review Outline	Review Plan
345	HSH Configuration Audit Plan	Audit Plan
346	HSH Software Design Assurance Plan	Software Plan
347	HSH Hardware Design Assurance Plan	Hardware Plan
348	HSH Supplier Quality Requirements Spec	Specification
349	HSH Acceptance Test Procedure Outline	Procedure
350	HSH Continuing Airworthiness Concept Outline	Maintenance Concept
351	HSH IP Element 0001 Occupied Survival Boundary	IP Description
352	HSH IP Element 0002 State-Dependent Selective Separation	IP Description
353	HSH IP Element 0003 Deterministic Safety Kernel	IP Description
354	HSH IP Element 0004 Six-Layer Survival Architecture	IP Description
355	HSH IP Element 0005 Eight-Support Impact Attenuation	IP Description
356	HSH IP Element 0006 Multi-Mode Recovery Chain	IP Description
357	HSH IP Element 0007 Partial Asymmetric Release Management	IP Description
358	HSH IP Element 0008 Residual Airframe Recovery Option	IP Description
359	HSH IP Element 0009 Immutable Emergency Event Record	IP Description
360	HSH Prior Art Search Plan Patent Landscape	IP Plan

Documents 361 – 390 — Verification, Certification, Compliance

No.	Document Name	Document Type
361	HSH Freedom-to-Operate Assessment Outline	IP Plan
362	HSH Claim Drafting Guidance Package Outline	IP Plan
363	HSH Licensing Data Room Contents List	Licensing Package
364	HSH Licensee Due Diligence Questionnaire	Licensing Package
365	HSH Strategic Licensing Path Description	Licensing Package
366	HSH Technology Platform Family Concept Utility Rescue	Architecture Description
367	HSH NDA Template Technical Evaluation Scope	Legal Template
368	HSH Controlled Data Room Access Procedure	Procedure
369	HSH Term Sheet Outline Territory Field of Use	Licensing Package
370	HSH Engineering Development Agreement Outline	Licensing Package

No.	Document Name	Document Type
371	HSH Program Work Breakdown Structure Level 1-3	Program Plan
372	HSH Integrated Master Schedule Outline	Program Plan
373	HSH Cost Estimating Framework Outline	Program Plan
374	HSH Resource and Skills Requirements Outline	Program Plan
375	HSH Make-Buy Strategy Outline	Program Plan
376	HSH Supplier Engagement Plan Outline	Program Plan
377	HSH Risk Register Initial Top Risks	Risk Register
378	HSH Opportunity Register Architecture IP	Register
379	HSH Stakeholder Communication Plan Outline	Management Plan
380	HSH Document Control and Numbering Procedure	Procedure
381	HSH Change Control Board Procedure	Procedure
382	HSH Lessons Learned Capture Procedure	Procedure
383	HSH Export Control Classification Outline	Compliance Spec
384	HSH Security Classification Handling Outline	Compliance Spec
385	HSH Environmental Regulatory Assessment Outline	Compliance Spec
386	HSH Occupational Health Safety Development Site	Safety Plan
387	HSH Prototype Build Philosophy Outline	Manufacturing Spec
388	HSH Tooling and Ground Support Equipment Outline	Manufacturing Spec
389	HSH Logistics Support Analysis Outline	Maintenance Concept
390	HSH Spares Provisioning Concept Outline	Maintenance Concept

Documents 391 – 420 — Manufacturing, Quality, Maintenance

No.	Document Name	Document Type
391	HSH Technical Publications Plan Outline	Documentation Spec
392	HSH Training System Development Outline	Training Spec
393	HSH Operator Transition Plan Outline	Program Plan
394	HSH Metrics and KPI Dashboard Definition	Management Plan
395	HSH Independent Review Board Charter Outline	Management Plan
396	HSH External Advisory Technical Panel Outline	Management Plan
397	HSH Academic Collaboration Framework Outline	Program Plan
398	HSH Public Disclosure and Marketing Claims Policy	Policy
399	HSH Document 00 to Document Tree Traceability	Traceability Matrix
400	HSH Master Document Tree Control Sheet 001-500	Control Sheet
401	HSH CAP-AF Structural Attachment Detailed Design	Design Description
402	HSH CAP-PAR Parachute Attachment Detailed Design	Design Description
403	HSH CAP-LG Landing Support Attachment Detailed Design	Design Description
404	HSH FUEL-ISO Isolation Valve Sequence Spec	Requirements Spec
405	HSH REC-LOC Locator Antenna Strategy Design	Design Description
406	HSH JET-CMD Command Path Independence Design	Design Description
407	HSH JET-PWR Segregated Power Architecture Design	Design Description
408	HSH JET-SNS Release State Sensor Redundancy Design	Design Description
409	HSH Capsule Internal Environment Post-Separation Spec	Requirements Spec
410	HSH Capsule Emergency Oxygen Duration Analysis	Analysis Report
411	HSH Capsule Emergency Communications Range Analysis	Analysis Report
412	HSH Capsule Locator Beacon Specification	Specification
413	HSH Capsule Post-Landing Lighting Specification	Specification
414	HSH Capsule Fire Suppression Agent Selection Study	Trade Study
415	HSH Capsule Smoke Detection Zoning Design	Design Description
416	HSH Capsule Thermal Protection Materials Study	Trade Study
417	HSH Capsule Waterproofing Sealing Design	Design Description
418	HSH Capsule Emergency Exit Marking and Lighting	Design Description
419	HSH Capsule Occupant Restraint Dynamic Analysis	Structural Analysis
420	HSH Capsule Seat Energy Absorber Stroke Analysis	Structural Analysis

Documents 421 – 450 — IP, Prior Art, Licensing Support

No.	Document Name	Document Type
421	HSH Eight-Support Load Distribution Optimization	Analysis Report
422	HSH Landing Support Soft Soil Penetration Study	Analysis Report
423	HSH Landing Support Slope Landing Stability Study	Analysis Report
424	HSH Landing Support Combined Load Cases Matrix	Structural Analysis
425	HSH Main Rotor Blade Mass Estimate 980 kg Closure	Mass Report
426	HSH Tail Rotor Blade Mass Estimate 145 kg Closure	Mass Report

No.	Document Name	Document Type
427	HSH Engine Installed Mass Estimate 800 kg Each	Mass Report
428	HSH Main Gearbox Mass Estimate 750 kg Closure	Mass Report
429	HSH Usable Fuel Mass 2450 kg Mission Closure	Mass Report
430	HSH Separation Hardware Mass 1000 kg Closure	Mass Report
431	HSH Avionics Group Mass 1600 kg Closure	Mass Report
432	HSH Airframe Group Mass 5000 kg Closure	Mass Report
433	HSH Growth Reserve 2000 kg Allocation Rules	Mass Report
434	HSH CG Shift During Staged Separation Analysis	Mass Properties
435	HSH Inertia Change During Staged Separation Analysis	Mass Properties
436	HSH Post-Separation Capsule CG and Inertia Report	Mass Properties
437	HSH Rotor Gyroscopic Effects on Separation Study	Analysis Report
438	HSH Tail Rotor Separation Dynamics Study	Analysis Report
439	HSH Engine Separation Dynamics Study	Analysis Report
440	HSH Fuel Tank Separation Dynamics Study	Analysis Report
441	HSH Main Rotor Non-Universal Jettison Rationale	Analysis Report
442	HSH Clearance Envelope Capsule vs Released Hardware	Analysis Report
443	HSH Parachute vs Released Hardware Collision Study	Safety Analysis
444	HSH Minimum Recovery Altitude Definition Analysis	Analysis Report
445	HSH Maximum Dynamic Pressure Separation Limit	Requirements Spec
446	HSH Attitude Rate Limits for Safe Separation	Requirements Spec
447	HSH Rotor State Criteria for Separation Authorization	Requirements Spec
448	HSH Fire State Criteria for Separation Authorization	Requirements Spec
449	HSH Structural Failure State Detection Criteria	Requirements Spec
450	HSH Sensor Disagreement Handling Logic Spec	Requirements Spec

Documents 451 – 475 — Program Management, Risk, Configuration

No.	Document Name	Document Type
451	HSH Power Availability Criteria for Release Recovery	Requirements Spec
452	HSH Communication Loss Safe State Definition	Requirements Spec
453	HSH Manual Inhibit Path Independence Verification	Safety Analysis
454	HSH Automatic Path Independence Verification	Safety Analysis
455	HSH Common Wiring Route Avoidance Design Rules	Design Criteria
456	HSH Common Power Bus Avoidance Design Rules	Design Criteria
457	HSH Common Structure Vulnerability Assessment	Safety Analysis
458	HSH Common Software Mode Avoidance Rules	Software Spec
459	HSH Fire Zone Segregation Design Rules	Design Criteria
460	HSH EMI Lightning Common Cause Mitigation	Design Description
461	HSH Manufacturing Process Spec Capsule CFRP	Manufacturing Spec
462	HSH Manufacturing Process Spec Airframe	Manufacturing Spec
463	HSH Assembly Sequence Capsule into Airframe	Manufacturing Spec
464	HSH Separation Mechanism Installation Procedure	Procedure
465	HSH Functional Check Procedure Separation System	Procedure
466	HSH Functional Check Procedure Safety Kernel	Procedure
467	HSH Functional Check Procedure Capsule Systems	Procedure
468	HSH Ground Run Procedure Propulsion Rotor	Procedure
469	HSH Pre-Flight Inspection Emergency Systems	Procedure
470	HSH Post-Event Inspection and Data Recovery	Procedure
471	HSH Maintenance Manual Chapter Emergency Systems Outline	Documentation Spec
472	HSH Illustrated Parts Catalog Emergency Hardware Outline	Documentation Spec
473	HSH Troubleshooting Guide Separation System Outline	Documentation Spec
474	HSH Service Bulletin Process Outline	Procedure
475	HSH Airworthiness Directive Response Process Outline	Procedure

Documents 476 – 500 — Digital Thread, Sustainability, Freeze, Sign-Off

No.	Document Name	Document Type
476	HSH Reliability Prediction Separation Mechanisms	Reliability Analysis
477	HSH Reliability Prediction Capsule Systems	Reliability Analysis
478	HSH Maintainability Analysis Emergency Systems	Maintainability Analysis
479	HSH Human Factors Assessment Emergency Controls	Human Factors
480	HSH Crew Workload Assessment Emergency Sequence	Human Factors
481	HSH Passenger Briefing Emergency Capsule Concept	Training Spec
482	HSH Rescue Services Interface Guidance Outline	Operational Spec

No.	Document Name	Document Type
483	HSH Accident Investigation Data Preservation Spec	Requirements Spec
484	HSH Simulation Validation Correlation Plan	V&V; Plan
485	HSH Model-Based Systems Engineering Approach Outline	Architecture Description
486	HSH Digital Thread Configuration Data Strategy	Management Plan
487	HSH Cybersecurity Considerations Avionics Outline	Security Spec
488	HSH Supply Chain Risk Assessment Outline	Risk Register
489	HSH Obsolescence Management Plan Outline	Management Plan
490	HSH End-of-Life and Disposal Concept Outline	Environmental Spec
491	HSH Sustainability Materials and Processes Outline	Environmental Spec
492	HSH Variant Derivative Control Procedure Outline	Procedure
493	HSH Single Main-Rotor Baseline Freeze Criteria Rev19	Checklist
494	HSH Open Items Closure Tracking Register (incl. OI-16 Airbags OI-17 Foam Seats)	Register
495	HSH Critical Path to First Ground Demonstration	Program Plan
496	HSH Critical Path to First Flight Development	Program Plan
497	HSH Licensing Readiness Checklist	Checklist
498	HSH Document Tree Completeness Audit Sheet	Audit Sheet
499	HSH Cross-Reference Matrix Doc00 Rev19 to Tree 001-500	Traceability Matrix
500	HSH Master Document Tree Sign-Off Record Rev19	Control Sheet

END OF HSH-00 REV19 MASTER ENGINEERING DOCUMENT TREE (001–500)

Aligned to HSH-00 Rev19 | 16 airbags | ≥50 cm foam seats | 3+2 canopies | 8 landing supports | Roll-cage docs 115–116 | Staged S0–S5 | Founder:
Amirbahador Atai

Continuous sequential numbering 1–500 with no gaps | Document Type shown for every entry